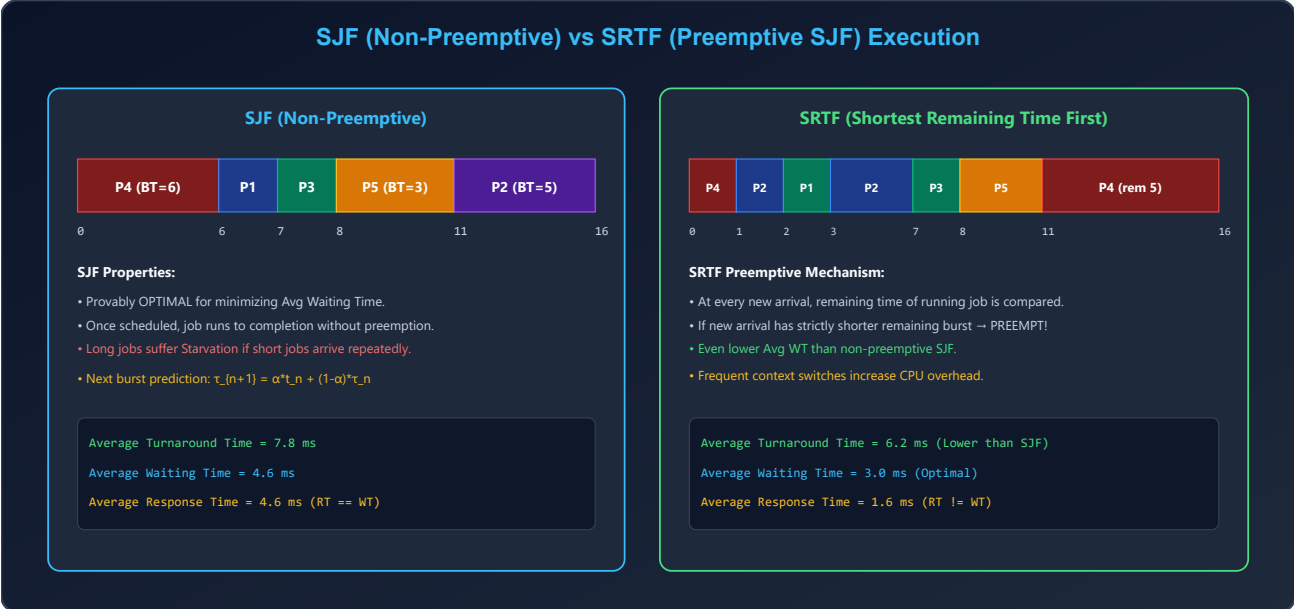


Module 06: SJF (Non-Preemptive) & SRTF (Preemptive SJF) Execution

1. Execution Comparison & Timelines



2. Exponential Smoothing Burst Prediction

$$\tau_{n+1} = \alpha t_n + (1 - \alpha) \tau_n$$

- t_n : Actual length of n-th CPU burst
- τ_n : Predicted length of n-th CPU burst
- α : Smoothing weight factor ($0 \leq \alpha \leq 1$)

3. Summary Comparison

Metric	Non-Preemptive SJF	Preemptive SJF (SRTF)
Preemption	No	Yes (At each arrival)
Avg WT	4.6 ms	3.0 ms (Lower / Superior)
Response Time	RT == WT	Significantly faster response
Context Switch Overhead	Low	High (Frequent preemption)